Xirui Li

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Research interests

My research interests lie in generative models in computer vision. Now I delve into diffusion models and work on their application in video generation and editing. I also have a background in point cloud object detection, graph representation learning, and 3D reconstruction.

Education

Shanghai Jiao Tong University Shanghai, China
MA in Computer Science September 2022 – March 2025
GPA: 3.92.

Shanghai Jiao Tong University Shanghai, China
BA in Computer Science September 2018 – June 2022
GPA: 3.84.

Publications

VidToMe: Video Token Merging for Zero-Shot Video Editing
Xirui Li, Chao Ma, Xiaokang Yang, Ming-Hsuan Yang.
CVPR, 2024. [Webpage] [Arxiv]

Frame Fusion with Vehicle Motion Prediction for 3D Object Detection
Xirui Li, Feng Wang, Naiyan Wang, Chao Ma.
ICRA, 2024. [Arxiv]

Experience

Shanghai Jiao Tong University - Vision and Learning Group
Mentors: Prof. Chao Ma September 2022 – Present
Delve into diffusion models and focus on diffusion-based video creation.
Propose a zero-shot video editing method based on token merging.

TuSimple - Algorithm Department
Mentors: Dr. Feng Wang February 2022 – June 2022
Explore the usage of motion models in 3D LiDAR object detection.
Develop a multi-frame fusion algorithm to improve point cloud vehicle detection performance.

Skills

Programming
Proficient in Python (Pytorch).
Familiar with C / C++.

Languages
English (fluent), Chinese (native)